



FEU INSTITUTE OF TECHNOLOGY

COMPUTER SCIENCE DEPARTMENT

CS0019

MODELING AND SIMULATION

TECHNICAL ASSESSMENT

2

Network Packet Queue Simulation in MATLAB

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Section:	TS31
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I. INTENDED LEARNING OUTCOME/S OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Identify the entities, resource, queue rule, system boundary, and events in a simple network simulation.
- Create a MATLAB simulation that computes packet start, waiting, and finish times.
- Compare simulation results after changing a packet arrival time and explain idle time.

II. BACKGROUND INFORMATION

A router forwards packets through one outgoing connection. A packet may contain only part of a larger message. For this activity, each packet has arrived in full before it joins the forwarding queue. The router transmits one whole packet at a time in first come, first served (FCFS) order. Packets arriving while the connection is busy wait in the queue.

The simulation changes at distinct events: packet arrival, transmission start, and transmission completion. Times are measured in milliseconds. The connection is idle when no packet is being transmitted.

AI classification: AI-Assisted. You may use AI to explain discrete event simulation, FCFS, or MATLAB syntax, to diagnose an error, and to suggest checks for your results. You must run and understand your own script. Do not submit unchecked AI-generated code, results, diagrams, or answers as your own, and do not claim tests you did not run.

III. LABORATORY ACTIVITY (100 POINTS)

Use MATLAB to model the following three packets. Packet names stay attached to their arrival and transmission times throughout both runs.

Packet	Arrival time (ms)	Transmission time (ms)
A	0	4
B	1	2
C	3	1

Part A. Define the model

Identify the entities, attributes, resource, state variables, events, FCFS queue rule, and system boundary. Draw a small flowchart showing what happens when a packet arrives and when a transmission ends.

Part B. Simulate the original arrivals in MATLAB

Write a MATLAB .m script using input arrays and a loop. Calculate the transmission start time, waiting time, and finish time for each packet. Start time is the later of its arrival and the previous packet's finish; waiting time is start minus arrival; finish time is start plus transmission time. Display all inputs and calculated values in a MATLAB table, followed by the average waiting time. Derive outputs from the input arrays; do not hard-code the calculated results.

Include screenshots of your output, make sure to show your name in the screenshots.

Part C. Change one arrival time

Change Packet C's arrival time from 3 ms to 7 ms. Run the script again, show the revised table and average waiting time, and explain in 2–3 sentences how Packet C's wait changes and when the connection is idle. Take a screenshots of the new outputs.

Part D. Questions to answer

1. What are the entities, resource, and events in your simulation?
2. Why does Packet B wait in the original scenario?
3. Does Packet A finishing transmission mean the destination has the complete message? Explain.
4. When is the outgoing connection idle in the revised scenario?
5. Why is this a discrete event simulation even though the MATLAB script uses a loop?

Submission

Submit the following files:

- .m file of Matlab source code
- the model description and flowchart, screenshots or exported output tables for both runs, answers to the five questions (this document in PDF)
- AI declaration form

IV. RUBRICS

Criterion	Points
Model description and flowchart correctly identify the system	15
MATLAB script uses input arrays and calculates packet times correctly	30
Original output and average waiting time are correct	15
Revised output and explanation of idle time are correct	15
Answers show understanding of events, queues, and packets	20
Submission is complete and AI use is declared	5
TOTAL	100

PART A. (Model and Flowchart)

Entities: Packets A, B, and C.

Attributes: Each packet has a packet name, arrival time, and transmission time. The simulation also calculates the start time, waiting time, and finishing time.

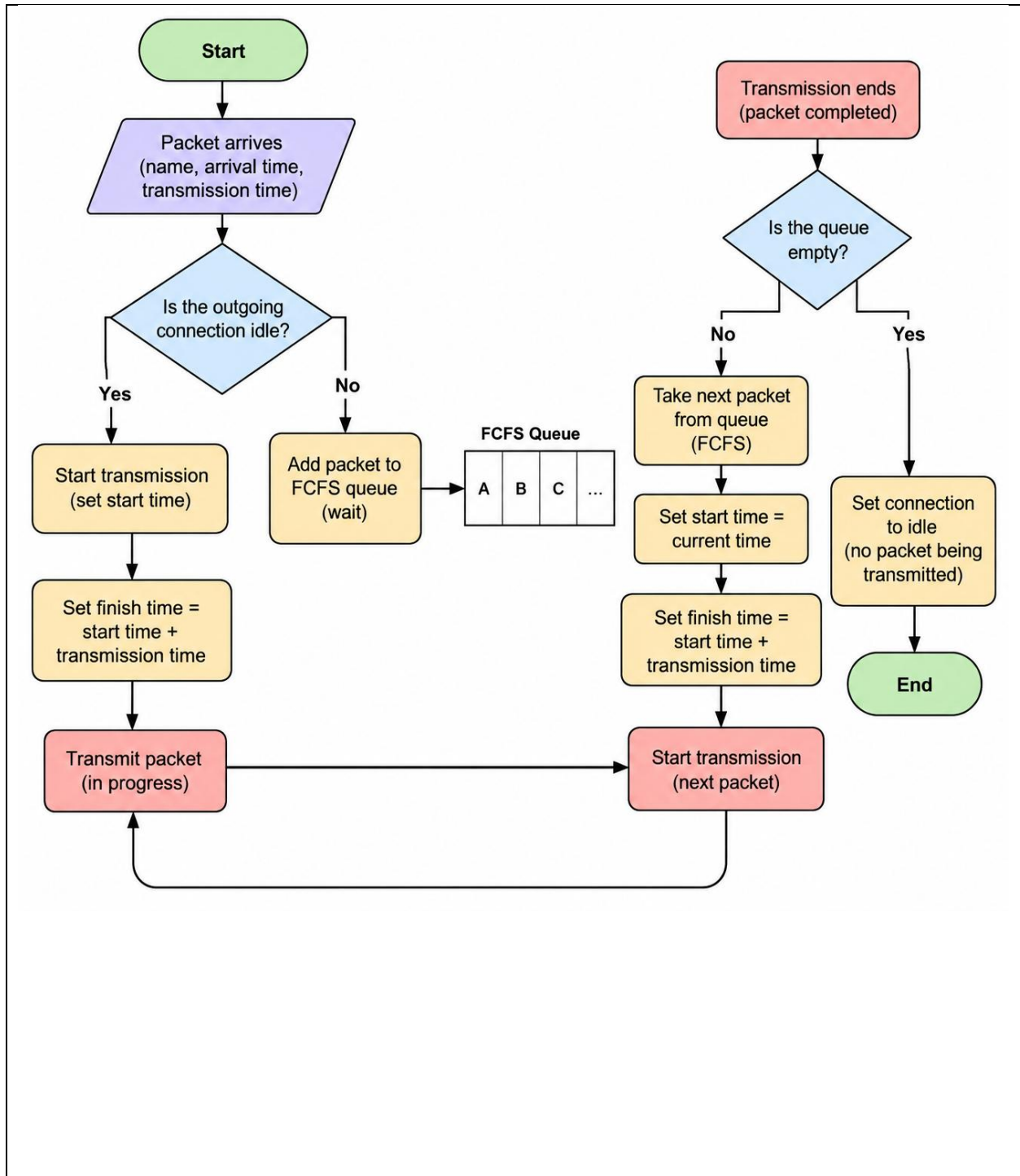
Resource: The router's single outgoing connection, which can transmit only one packet at a time.

State Variables: The status of the outgoing connection (busy or idle), the packets waiting in the queue, and the finish time of the packet currently being transmitted.

Events: Packet arrival, transmission start, and transmission completion.

FCFS Queue Rule: First Come, First Served (FCFS). Packets are transmitted according to their order of arrival. If the outgoing connection is busy, an arriving packet waits in the queue.

System Boundary: The system begins when a packet arrives at the router. It includes waiting in the queue and transmission through the outgoing connection. The packet leaves the system when its transmission is completed.



PART B. (Screenshots)

```
untitled * × +
1  clc;
2  clear;
3
4  % Student information
5  studentName = "John Kizon Guo, Dawn Andrei Pamesa";
6
7  % Input data
8  packet = ["A"; "B"; "C"];
9  arrivalTime = [0; 1; 3];
10 transmissionTime = [4; 2; 1];
11
12 % Number of packets
13 n = length(packet);
14
15 % Initialize calculated values
16 startTime = zeros(n,1);
17 waitingTime = zeros(n,1);
18 finishTime = zeros(n,1);
19
20 % FCFS simulation
21 for i = 1:n
22
23     if i == 1
24         % First packet can start when it arrives
25         startTime(i) = arrivalTime(i);
26     else
27         % Start at arrival or when previous packet finishes,
28         % whichever is later
29         startTime(i) = max(arrivalTime(i), finishTime(i-1));
30     end
31 end
```

```
untitled * x +
25     startTime(i) = arrivalTime(i);
26     else
27         % Start at arrival or when previous packet finishes,
28         % whichever is later
29         startTime(i) = max(arrivalTime(i), finishTime(i-1));
30     end
31
32     % Calculate waiting and finish times
33     waitingTime(i) = startTime(i) - arrivalTime(i);
34     finishTime(i) = startTime(i) + transmissionTime(i);
35
36 end
37
38 % Calculate average waiting time
39 averageWaitingTime = mean(waitingTime);
40
41 % Create output table
42 results = table(packet, arrivalTime, transmissionTime, ...
43     startTime, waitingTime, finishTime, ...
44     'VariableNames', {'Packet', 'ArrivalTime_ms', ...
45     'TransmissionTime_ms', 'StartTime_ms', ...
46     'WaitingTime_ms', 'FinishTime_ms'});
47
48 % Display results
49 fprintf('Student Name: %s\n\n', studentName);
50 fprintf('=== ORIGINAL PACKET QUEUE SIMULATION ===\n\n');
51
52 disp(results);
53
54 fprintf('Average Waiting Time: %.2f ms\n', averageWaitingTime);
```

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=== ORIGINAL PACKET QUEUE SIMULATION ===

Packet	ArrivalTime_ms	TransmissionTime_ms	StartTime_ms	WaitingTime_ms	FinishTime_ms
"A"	0	4	0	0	4
"B"	1	2	4	3	6
"C"	3	1	6	3	7

Average Waiting Time: 2.00 ms

>>

PART C. (Screenshots)

The screenshot shows a MATLAB script and its execution output. The script defines arrival and transmission times for three packets (A, B, C) and simulates two scenarios: original arrivals and a revised scenario where Packet C arrives at 7 ms instead of 3 ms. The Command Window displays the results of these simulations, including tables of packet times and idle periods.

```

13 arrivalTime = [0; 1; 3];
14 transmissionTime = [4; 2; 1];
15
16 fprintf('Student Name: %s\n', studentName);
17
18 %% Part B. Original arrivals
19 fprintf('=== ORIGINAL PACKET QUEUE SIMULATION ===\n\n');
20 simulateQueue(packet, arrivalTime, transmissionTime);
21
22 %% Part C. Packet C arrives at 7 ms instead of 3 ms
23 revisedArrival = arrivalTime;
24 revisedArrival(packet == "C") = 7;
25
26 fprintf('\n=== REVISED PACKET QUEUE SIMULATION (Packet C arrives at 7 ms) ===\n\n');
27 [startTime, finishTime] = simulateQueue(packet, revisedArrival, transmissionTime);
28
29 fprintf('\nIdle periods of the outgoing connection:\n');
30 for i = 2:numel(packet)
31     if startTime(i) > finishTime(i-1)
32         fprintf(' %d ms to %d ms (before Packet %s)\n', finishTime(i-1), startTime(i), packet(i));
33     end
34 end
35 fprintf(' after %d ms (no packets left)\n', finishTime(end));
36

```

Command Window Output:

```

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=== ORIGINAL PACKET QUEUE SIMULATION ===

Packet   ArrivalTime_ms   TransmissionTime_ms   StartTime_ms   WaitingTime_ms   FinishTime_ms
-----
"A"       0                 4                   0              0                4
"B"       1                 2                   4              3                6
"C"       3                 1                   6              3                7

Average Waiting Time: 2.00 ms

=== REVISED PACKET QUEUE SIMULATION (Packet C arrives at 7 ms) ===

Packet   ArrivalTime_ms   TransmissionTime_ms   StartTime_ms   WaitingTime_ms   FinishTime_ms
-----
"A"       0                 4                   0              0                4
"B"       1                 2                   4              3                6
"C"       7                 1                   7              0                8

Average Waiting Time: 1.00 ms

Idle periods of the outgoing connection:
 6 ms to 7 ms (before Packet C)
after 8 ms (no packets left)
>>

```

Explanation: Moving Packet C's arrival from 3 ms to 7 ms drops its waiting time from 3 ms to 0 ms. Packet B now finishes at 6 ms, so the outgoing connection is already free when Packet C arrives at 7 ms and it starts right away, which lowers the average waiting time from 2.00 ms to 1.00 ms. The connection is idle from 6 ms to 7 ms, between Packet B finishing and Packet C arriving, and again after Packet C finishes at 8 ms.

PART D. ANSWERS TO QUESTION

1. What are the entities, resource, and events in your simulation?

The entities are Packets A, B, and C. The resource is the router's single outgoing connection, which can transmit only one packet at a time. The events are packet arrival, transmission start, and transmission completion.

2. Why does Packet B wait in the original scenario?

Packet B arrives at 1 ms while Packet A is still being transmitted from 0 ms to 4 ms. Since the connection can only send one packet at a time and the queue is FCFS, Packet B waits in the queue until Packet A finishes at 4 ms, which is a waiting time of 3 ms.

3. Does Packet A finishing transmission mean the destination has the complete message? Explain.

No. A packet may contain only part of a larger message, so Packet A finishing only means that one packet has left the router through the outgoing connection. The destination has the complete message only after every packet of that message has been transmitted and has reached it. Our system boundary also ends at transmission completion, so the simulation does not model the packet's travel to the destination.

4. When is the outgoing connection idle in the revised scenario?

The connection is idle from 6 ms to 7 ms, after Packet B finishes and before Packet C arrives. It is also idle after 8 ms, once Packet C finishes and no packets are left.

5. Why is this a discrete event simulation even though the MATLAB script uses a loop?

The state of the system only changes at specific moments: when a packet arrives, when a transmission starts, and when a transmission ends. The loop does not step through time in fixed ticks. Each pass handles the events of one packet in FCFS order and jumps straight from one event time to the next, so nothing is simulated between events. The loop is just how the script goes through the event list.